H1N1 Vaccine

Uptake Prediction

Using Machine Learning to Predict Vaccination Decisions
During the 2009 H1N1 Influenza Pandemic

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Business Understanding — Overview

Context

During the 2009 H1N1 influenza pandemic, public health organizations faced the challenge of ensuring as many people as possible received the H1N1 vaccine.

Not everyone chose vaccination due to differences in:

- Health beliefs
- Demographic factors
- Access to healthcare

Understanding which groups are more or less likely to receive the vaccine can help design better vaccination campaigns and allocate resources more effectively.

Project Goal

Build an ML model to predict H1N1 vaccine uptake using demographics, health behaviors & opinions.

Value Delivered

Targeted awareness campaigns, improved vaccine rates, reduced hospitalisations & healthcare costs.

Stakeholders

Healthcare organizations, policymakers and public health officials.

Business Problem & Objectives

Business Problem

- Cannot easily identify groups with low vaccination uptake
- Campaigns may not effectively reach hesitant populations
- Healthcare resources may not be allocated efficiently during outbreaks
- Misconceptions about vaccine safety continue to influence decisions

Objectives

1. Build a classification model to predict H1N1 vaccine receipt
2. Evaluate using accuracy, precision, recall, F1 and ROC-AUC
3. Identify the most important factors influencing decisions
4. Provide insights for more effective vaccination campaigns
5. Support policymakers in improving vaccination coverage

Data Understanding

26,707	38	35	21.4%
Respondents	Features	Predictor Variables	Vaccinated (minority class)

Key Feature Categories

Demographics:

Age group, Sex, Race, Education, Income, Marital status

Health Behaviors:

Antiviral meds, Avoidance, Face mask, Hand washing, Large gatherings

Health Background:

Chronic condition, Child under 6mo, Health worker, Health insurance

Vaccine Opinions:

H1N1 concern, Knowledge, Perceived risk, Vaccine effectiveness beliefs

Doctor's Role:

Doctor recommendation for H1N1 and seasonal flu vaccines

Data Preprocessing Pipeline

1

Train-Test Split

80/20 split (21,365 train / 5,342 test) with stratification to preserve class proportions. Applied BEFORE any preprocessing to prevent data leakage.

2

Handling Missing Values

Numerical columns: filled with median values. Categorical columns: filled with mode. Health insurance had the most missing data (9,849 entries in training set).

3

One-Hot Encoding

Categorical variables converted to binary indicator columns using pd.get_dummies with drop_first=True to avoid multicollinearity (dummy variable trap).

4

Standard Scaling

StandardScaler applied to normalise all numerical features so no single feature dominates due to scale differences. Fit on training data only.

5

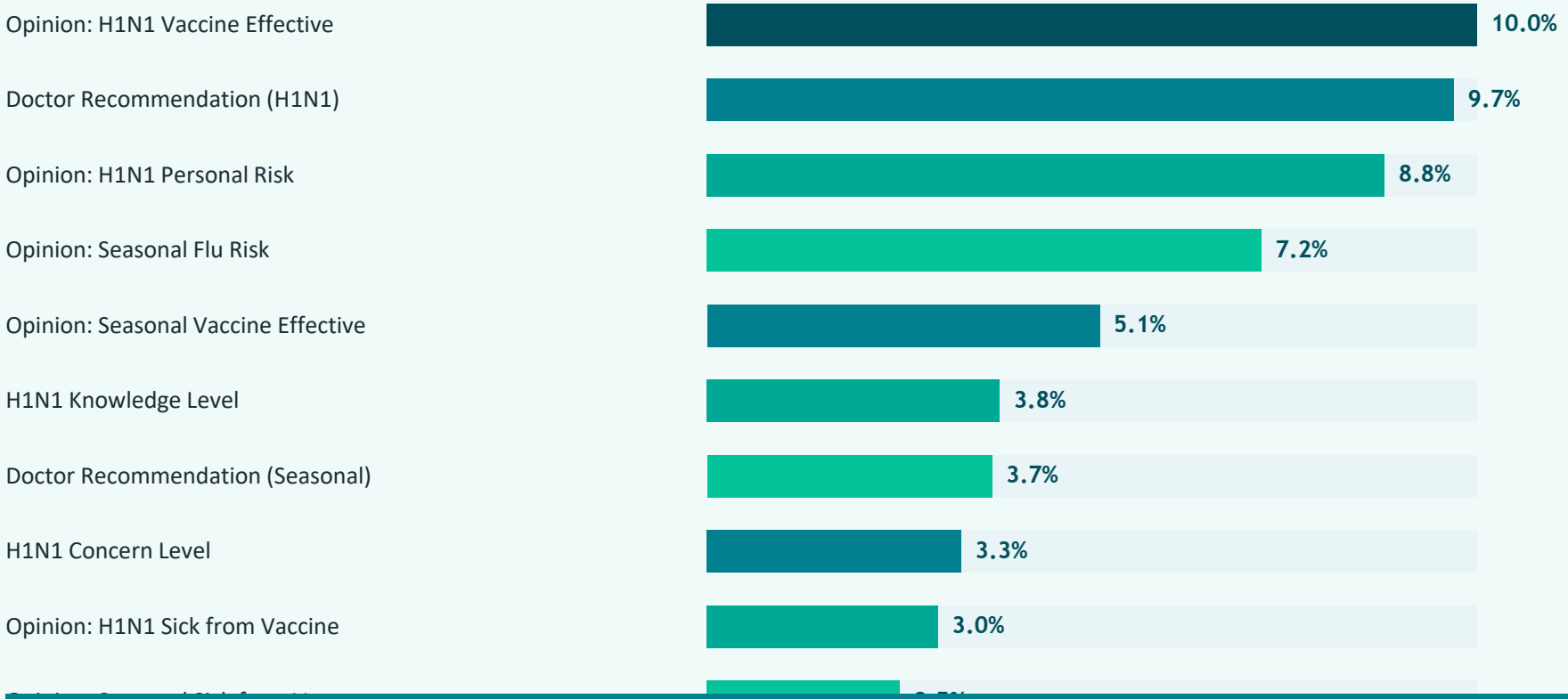
SMOTE Balancing

Dataset was highly imbalanced (only ~21% vaccinated). SMOTE applied to training data to generate synthetic minority class samples, improving recall and F1-score.

Model Training & Results

	Accuracy	Precision	Recall	F1	ROC-AUC
Ridge Logistic Regression *L2 regularization prevents overfitting* High recall — good at identifying vaccinated individuals. Moderate precision.	77.7%	48.3%	72.7%	0.580	0.826
Lasso Logistic Regression *L1 regularization performs feature selection* Nearly identical to Ridge. Marginally better F1.	77.7%	48.3%	72.7%	0.581	0.826
Random Forest *200 estimators, ensemble tree model* High accuracy & precision but lower recall — misses more vaccinated individuals.	83.8%	68.4%	44.3%	0.538	0.822
Gradient Boosting ★ Best *Sequential boosting, best overall balance* Best AUC & F1 balance. Recommended for deployment.	83.6%	65.1%	49.4%	0.562	0.828

Feature Importance — Top Predictors



💡 Key Insight: Personal opinions about vaccine effectiveness & risk, plus doctor recommendations, are the strongest drivers of H1N1 vaccination decisions.

Model Comparison — Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Ridge LR	77.67%	48.30%	72.69%	0.580	0.826
Lasso LR	77.69%	48.33%	72.69%	0.581	0.826
Random Forest	83.83%	68.44%	44.32%	0.538	0.822
Gradient Boosting	83.62%	65.08%	49.43%	0.562	0.828

Recommended Model

Gradient Boosting Classifier

Achieves the best overall balance: highest AUC (0.828), strong accuracy (83.6%), best F1 among tree models (0.562) and superior precision-recall tradeoff. Ideal for deployment in public health decision-making.

Conclusions & Recommendations

Conclusions

- ✓ The Gradient Boosting Classifier performed best overall (AUC 0.828, Accuracy 83.6%)
- ✓ Personal opinions about vaccine effectiveness & risk were the #1 drivers of vaccination decisions
- ✓ Doctor recommendations are highly influential — healthcare provider outreach is critical
- ✓ H1N1 knowledge and concern levels significantly impact uptake decisions
- ✓ Class imbalance was successfully addressed using SMOTE, improving minority class detection

Recommendations for Public Health Organisations

🎯 Use the Gradient Boosting model to identify and target hesitant populations

📢 Launch awareness campaigns focused on vaccine safety & personal risk information

👩‍⚕️ Engage doctors to proactively recommend H1N1 vaccines to at-risk patients

📊 Allocate healthcare resources efficiently using model predictions during outbreaks